Iliad – June 29 Notes

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1 Overview

[Additional plots and context can be found at here.]

- **Spectra of weight matrices (§2):** on a single Pythia-70m training run we track each weight matrix's $W^\top W$ spectrum from a Marchenko–Pastur bulk through BBP spikes to a power-law tail and relate the fitted exponent α to held-out loss (§2.1), argue the heavy tail is built from learned correlations rather than heavy weight entries (§2.2), and flag why the further step to a genuinely sparse / localized representation does not yet follow (§2.3).
- **eNTK experiments (§3):** extending Lin's result that eNTK eigendirections track features, we follow how those directions rotate over Pythia-70m training — a burst that precedes the heavy-tail onset of §2 (§3.1) — and test whether deflating the kernel's trivial background isolates the feature, cleanly in a modular toy and as a tradeoff in Gemma (§3.2).

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2 Spectra of weight matrices

Since Charles already wrote up some of his more more theoretical hopes/ideas (blog post), I'm just including here the more empirical side – to add additional context, especially for the parts of the story where empirical evidence in relevant examples is either weak or somewhat in tension with the picture as laid out there. As a reminder, the proposal is that a power-law weight-spectrum exponent α — the slope of the power-law tail fit to the empirical spectral density (ESD) of $W^\top W$, i.e. $\rho(\lambda) \sim \lambda^{-\alpha}$ for large λ, with smaller α a heavier tail (and $\alpha = 2$ the finite-variance boundary)

— interpolates between true sparsity and Gaussian density, and spectrum-as-compressibility is a signature of the network implementing Occam's razor, in a way that could yield a possible mechanism for inductive bias toward sparse/factored representations. Additionally, much of this is based on prior work (Martin and Mahoney; Prakash–Martin); in particular, several of the empirical results below are focused on sanity checks/reproducing parts of that work, rather than adding to it – I'll try to flag where that's the case.

I take the post's main claims in turn, noting for each whether the empirical picture in the examples I have reproduces it, agrees with it, sits in tension with it, or is so far untested.

2.1 Spectra develop power-law tails over training, and α tracks quality

Claim. Weight-matrix spectra evolve over training from a Marchenko–Pastur bulk, through BBP spikes, to a broad heavy-tailed (power-law) ESD, and the fitted exponent α predicts generalization without train/test data.

Prior evidence. This is broadly supported, though the strongest existing results are only mostly partial. Martin and Mahoney describe these phases [1], but primarily demonstrate them by surveying pretrained production models of various sizes — not dynamically along a single training run. The prior work that genuinely tracks over training is narrower: in the grokking setting Prakash–Martin follow α over training steps through the phases, and beren's LessWrong post [2] tracks weights, activations, and gradients developing heavier-than-Gaussian (logistic) tails across Pythia checkpoints, which our own over-training fits broadly confirm.[1] On the α-as-generalization-metric side, the weightwatcher paper+website [3] tie α to test accuracy as a data-free quality metric, again across pretrained models of varying depth and size rather than over training (and even there α beats other, simpler metrics[2] only marginally / architecture-dependently); the value $\alpha = 2$ marks the finite-variance / "ideal-learning" point, with well-trained layers typically fitting $\alpha \in [2, 4]$.

New work. We mostly reproduce the beren / Martin and Mahoney weight-spectrum picture on Pythia-70m, across training, and with more careful analysis. In particular, we track the various phase sequences along a single training run broken up by layer and matrix type[3], verify some of the underlying fits from the beren blog, and also track additional relevant objects. Main findings:

- Across the 143k training steps steps, every $X = W^\top W$ block (roughly) goes from pure MP → MP + spikes → heavy tail + MP (with some moving → just heavy tail), with most of the motion in steps ∼500–32k; see Figure 1.

[1] beren's vocabulary is "logistic" / "power-law" / "heavy-tailed"; he never invokes α-stable or infinite variance, and explicitly brackets the weight entries "between the logistic e^{-x} and the Gaussian e^{-x^2}" — i.e. finite-variance. Our entry fits (second claim) confirm this for the bulk; the one place we go past him is the deep-layer QKV entries, which become genuinely heavier than his logistic ceiling.

[2] The "simpler metrics" are norm-based *scale* measures (not model size or parameter count): principally the *log spectral norm* — $\langle \log \lambda_{\max} \rangle$, the log of the top $W^\top W$ eigenvalue $\lambda_{\max}$ averaged over the network's blocks. The contrast is scale (how big the weights are) versus shape (α, the tail slope, which is scale-invariant).

[3] attention QKV (the fused query/key/value projection), attention-out (the attention output projection W_O), MLP-in (the $d \to 4d$ up-projection), and MLP-out (the $4d \to d$ down-projection)

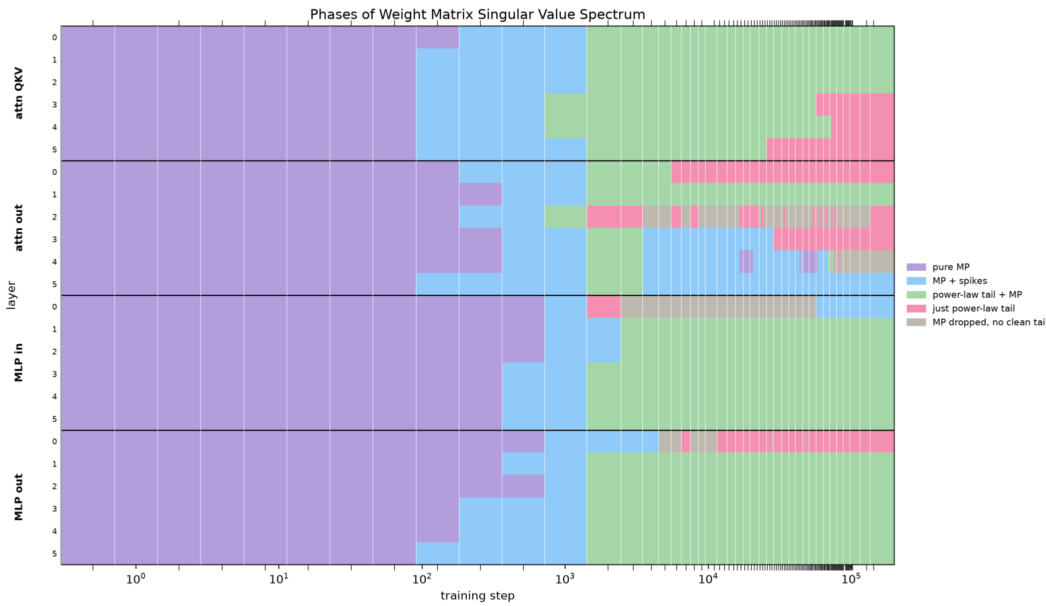

Figure 1: Phase of each weight matrix's singular-value spectrum over training (Pythia-70m). Top: the 24 layer×block matrices (attn QKV, attn out, MLP in, MLP out, each over six layers), classified per checkpoint as pure MP → MP + spikes → heavy tail + MP → just heavy tail (the last = MP bulk fully dissolved — a shape call, independent of the exact α). Several blocks reach "just heavy tail" late in training, the deep QKV most persistently (their α-trajectories are in Figure 2). Interactive version in the live dashboard.

- *Which blocks go heavy, and when.* The heaviness concentrates in the deep-layer QKV: only the deepest QKV blocks fall below the finite-variance boundary $\alpha = 2$, while every other block settles in or above the healthy band. It is also a late event — after an early, universal drop into the healthy range over the first couple of thousand steps, the early-layer QKV re-lighten while only the deep QKV keep collapsing into genuinely heavy territory, and that only in the later stretch of training; see Figure 2.

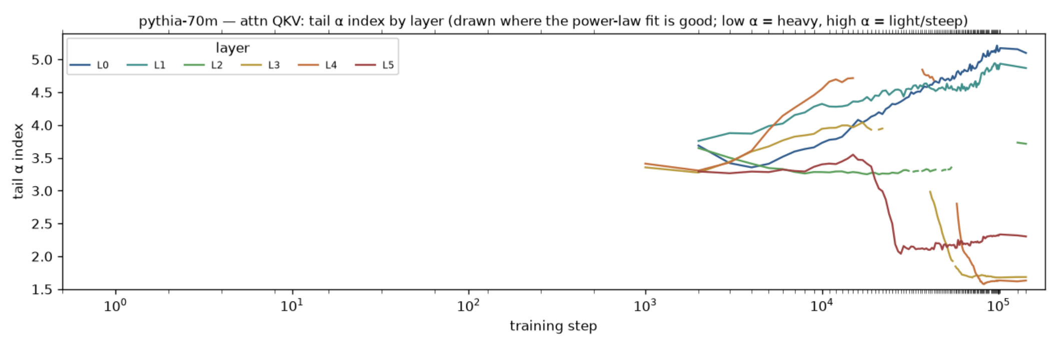

Figure 2: Heavy-tail exponent $\alpha(t)$ for the attention QKV blocks, coloured by layer (Pythia-70m; each curve is drawn only once that matrix has developed a heavy tail; lower α = heavier). The bifurcation is the story: the early-layer QKV re-lighten over training (toward $\alpha \sim 5$), while the deep-layer QKV (L3/L4/L5) fall below the finite-variance boundary $\alpha = 2$ late in training.

- *α and generalization prediction.* Our eval signals are held-out cross-entropy loss and LAMBADA 0-shot accuracy — the fraction of LAMBADA passages whose final word the model predicts correctly (greedy next-token match, no in-context examples; the passages are built so the last word needs the whole passage, not just the last sentence) — plotted against the phase transitions (Figure 3): globally, the loss flattens and accuracy lifts off as the MP bulk gives way to the heavy tail. But asked which spectral summary best follows the eval curves, the answer is not flattering to the tail exponent — the (log) spectral norm $\log \lambda_{\max}$ tracks them more closely than the mean α does (the weighted $\langle \alpha \log \lambda_{\max} \rangle$ follows only because it folds the norm back in). This may be partly an artifact: most blocks never develop a genuine power-law tail, so averaging α over blocks mixes a few real tail fits with many non-power-law ones. Either way, it is unclear how far the tail exponent α should be treated as the special quantity.

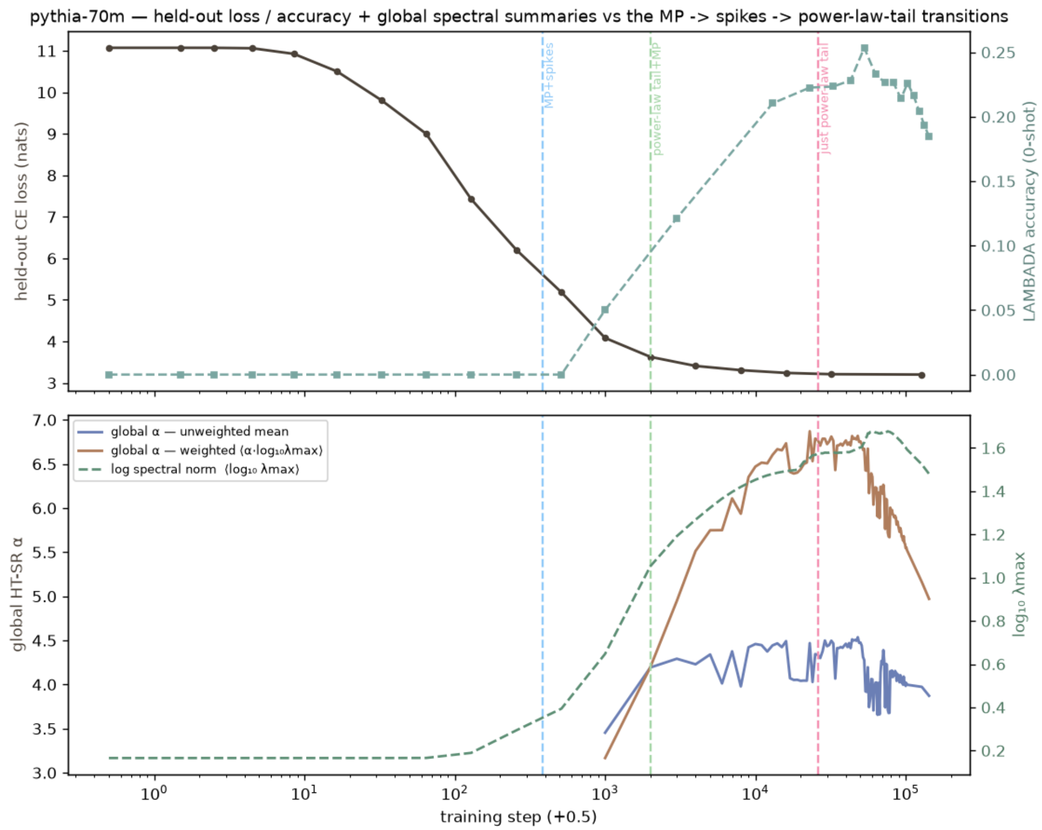

Figure 3: Held-out loss / accuracy and global spectral summaries vs the phase transitions (Pythia-70m). Top: held-out CE loss (left, nats) and LAMBADA 0-shot accuracy (right). Bottom: global HT-SR α — the unweighted mean and the Martin and Mahoney weighted $\langle \alpha \log_{10} \lambda_{\max} \rangle$ — with $\log_{10} \lambda_{\max}$. Vertical dashed lines mark the global MP + spikes, heavy tail + MP, and just-heavy-tail transitions; the loss has largely flattened by the heavy-tail onset.

Aside — anti-grokking. Beyond the four spectral phases above, we also try to track the additional late-training phases spelled out by Prakash and Martin [4, 5], including the late-stage generalization collapse ("anti-grokking," with α dropping back below 2). We reproduce the full grok→collapse on MNIST, but could not reproduce it for modular arithmetic — across several handfuls of configurations the network groks and holds with no collapse, so it appears to need Prakash and Martin's unpublished modular-addition recipe.[4]

[4]Separately, the modular-arithmetic weight *updates* show Lévy-flight-like bursts over training — the increment distribution ΔW going heavy-tailed at isolated transition steps, in the heavy-tailed-SGD-noise vein [6] (and the Wesley Erickson talk the blog points to), which we reproduce. This concerns the update dynamics, not the static entry distribution — which stays light-tailed, consistent with the blog. And in these settings, spikes pulling out of the MP bulk generally coincides with the onset of grokking.

2.2 The heavy tail is correlation-driven, not heavy-entry

Claim. Power laws form a generalized-CLT universality class — the heavy-tailed analogue of the Gaussian/CLT — and in NNs the heavy spectrum comes from correlations (light-tailed entries plus many small correlations), not from heavy-tailed entries; the post as it now stands explicitly rejects the Lévy-matrix-of-heavy-entries picture, though an earlier version instead attributed the heavy tail to heavy-tailed structure in W itself. The tests below are aimed at that initial, naive version — heavy spectrum because the entries of W are heavy — rather than the current correlation-driven statement, and serve to caveat it.

Prior evidence. Martin and Mahoney are explicit that heavy-tailed self-regularization is a claim about the *spectral* density: they only model the strongly-correlated W *as if* it were drawn from a heavy-tailed random-matrix class, with the weights heavy "correlation-wise, not element-wise" and the entries *not* i.i.d. heavy-tailed draws [1, 7]. But the picture is not clean: a separate line reports genuinely heavy-tailed / α-stable (Lévy) structure arising in training — in the stochastic-gradient noise [6] and in the SGD iterates / stationary weight distribution themselves [8, 9] (claims about the noise and the weights, not the ESD). So the most naive version — that the spectrum is heavy because the entries are heavy — is worth testing directly rather than assuming away. One caveat keeps even a negative result from closing the question: entry-heaviness is *basis-dependent* — an orthogonal rotation of W leaves the singular values (and α) untouched but mixes the entries toward Gaussian — so the weights could in principle be genuinely heavy-tailed in *some* basis while looking light in the storage basis. Whether such a basis exists is unclear, and from what we see unlikely, but it is the one loophole a storage-basis entry fit cannot close.[5]

New work. The most naive version — a heavy spectrum because W has heavy *entries* — plainly fails. The entries are too light: as iid draws they give Marchenko–Pastur, and the standard shuffle test[6] — which destroys correlations but preserves the entry distribution — collapses the tail (median α 3.5 $\to$ 20+, $\lambda_{\max}/\mathrm{MP} \to 1$) for 25 of 26 matrices. Tracking the entries directly (5-family fits, steps 0$\to$143k) mostly just confirms beren — every matrix starts Gaussian and drifts only mildly sub-Gaussian — the lone exception being the deep-layer QKV, which go genuinely α-stable (≈ 0.7–1.2), past his logistic ceiling.[7] So the entry-heaviness is not what makes the spectrum heavy: the tail lives in the learned correlations, consistent with its growing out of a Gaussian init — with the same caveat as above, that shuffling and entry-heaviness are both storage-basis notions, so this rules out heavy entries *as stored*, not a heavy basis reachable by an orthogonal rotation.

On the universality framing: the generalized CLT describes the power-law *shape*, but the mechanism here is correlation, not a sum of heavy entries — agreement with his picture, sharpened.

2.3 α as a compressibility proxy for sparse, factored representations

Claim. α is a smooth generalization of sparsity: below $\alpha = 2$ the best-k truncation error decays as a power law, so α dials between truly sparse and Gaussian-dense; and a heavy spectrum is read as implying sparse / factored / localized representations, hence inductive bias (an Occam's-razor signature).

[5] Our own entry fits (below) land in between: finite-variance in the bulk for almost every matrix, genuinely heavy (α-stable < 1) only in the deep QKV.

[6] The shuffle$\to$MP diagnostic is Martin and Mahoney's / Prakash–Martin's "correlation traps" [10]; we run their test and get their expected answer.

[7] Largely a sanity check on data already collected for the eNTK analysis. Robustness: McCulloch α-stable and kurtosis agree for the deep QKV (both heavy) but diverge for some MLP matrices (a Gaussian bulk with a few outlier weights, not a uniform heavy law); the QKV $\alpha < 1$ values are McCulloch point estimates pending a tail goodness-of-fit, and a whitening/ICA probe for independent heavy structure was inconclusive.

New work. The step that needs caveating is the leap from a compressible spectrum to a sparse, *localized* representation. This part is not yet well fleshed out, and — as in the previous subsection — the checks below mostly target the earlier, more concrete and naive versions of the claim rather than its current form. *[will fill in later]*

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3 eNTK experiments

Lin [11] shows that eNTK eigendirections track features: in modular-arithmetic toys and in Gemma-3-270M, the top eigenvectors of the empirical NTK line up with known features (Fourier modes; grammatical categories). We do some simple extensions/experiments in two directions

- *Dynamics in Language Models:* We track how the feature directions rotate over training in Pythia-70m, and how this corresponds to e.g. bulk spikes emerging in the spectral distribution for weight matricies as tracked above [8]
- *Centering:* We also test subtracting off the eNTK's "trivial" background — the constant term, the diagonal/self-term, and the single-token ($XX^\top$) activation structure (the El Karoui shells [12]) — to see if just the leftover "non-linear" portion tracks more interesting features.

3.1 Tracking over training

Early in training, a burst of eNTK eigenvector rotation immediately precedes the onset of heavy-tail development in the Pythia weight spectra of §2. A second, later rotation tracks the dissolution of the Gaussian (Marchenko–Pastur) bulk, but only for certain blocks and less cleanly — possibly noise.

3.2 Centering

In the modular toy, centering cleanly isolates the grokked sum-feature ($a + b$): invisible in the full kernel (which sees only the single-token band), it dominates the deflated ("non-linear") kernel across grokking. In Gemma it is a tradeoff — deflating the constant and top PCA directions drops the features PCA captures while keeping the eNTK's non-PCA edge, where it still beats a PCA baseline on a subset of grammar features.

References

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[2] beren. Basic facts about language models during training. `https://www.lesswrong.com/posts/2JJtxitp6nqu6ffak/basic-facts-about-language-models-during-training-1`, 2023. LessWrong.

[8] Note that for convenience, we replace the expensive matrix-valued residual kernel by a cheap scalar collapse. This is partially justified by Lin's results in Gemma, and partially by our own testing, showing that this collapse retains the features she finds in Gemma

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